

Materials and Methods

Coupling disease surveillance and epidemic dynamics

Our coupled behavioral-epidemiological model depicted in Figure 1 assumes the epidemic initially spreads among a naive population with a Susceptible-Infected-recovered scheme formalized by equation (S1)

$$\begin{aligned}\dot{S} &= -\beta S \frac{I}{N}, \\ \dot{I} &= \beta S \frac{I}{N} - \gamma I, \\ \dot{R} &= \gamma I,\end{aligned}\tag{S1}$$

where β corresponds to the transmission rate, and γ is the recovery rate. We couple the disease progression model with a surveillance model formulated using a probabilistic approach to compute the expected time required to identify at least a single infected individual using power calculations. Assuming a homogeneous mixing population and a random sampling strategy, we compute the cumulative probability of detecting at least one infectious individual. The proposed disease detection model incorporates the dynamic progression of the epidemic into the daily probability of detecting infectious individuals. The probability of detecting at least a single positive case by time t , for a disease prevalence level of ϕ_s on day s , for surveillance effort of N_s daily tested individuals, is given by

$$\mathbb{P}_{det} = 1 - \prod_{s=1}^t (1 - \phi_s)^{N_s}.\tag{S2}$$

Finally, we let the disease detection time (τ_{det}) to be reached at the time at least a single infectious individual is detected with at least 95% confidence, $\mathbb{P}_{det} = 0.95$. The detection time determines the disease prevalence threshold (P^*) at which health emergency declarations are issued. Note that P^* is not arbitrarily selected; instead it depends on the preselected surveillance effort.

Disease surveillance triggers interventions, risk information spread and adaptive human behavior

Upon health emergency declaration, we assume the onset of centralized intervention quarantining a fraction ϕ newly infected individuals (Q), and that the susceptible population is divided into the